

ABSTRACT

ProComp introduces a problem-driven multi-agent review paradigm that replaces unstable absolute scoring with structured analysis and comparative ranking. By integrating significance assessment, methodological validation, and contribution synthesis, it delivers more coherent and interpretable evaluations

RESEARCH QUESTIONS

- How effectively can specialized agents capture key aspects of a paper, such as problem significance, methodology rigor, novelty, and results impact?
- How reliably do LLM review scores preserve the ordinal structure of paper quality groups through discriminative scoring behavior?
- How well does the multi-agent evaluation system is better while compared with direct prompting and existing baseline methods?

CONCLUSION & FUTURE WORK

- ProComp's specialized agents consistently detect quality differences across dimensions, producing clear, human-aligned distinctions. This shows that structured evaluation is more rigorous and interpretable than one-shot scoring.
- Building on these findings, our next steps involve expanding ProComp's domain range, developing more adaptive agent behaviors, and evaluating how structured multi-agent scoring can enhance or streamline human peer-review pipelines.

REFERENCES

- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824–24837.
- Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)* (pp. 1–22). Association for Computing Machinery. <https://doi.org/10.1145/3586183.3606763> (doi.org in Bing)

METHODOLOGY

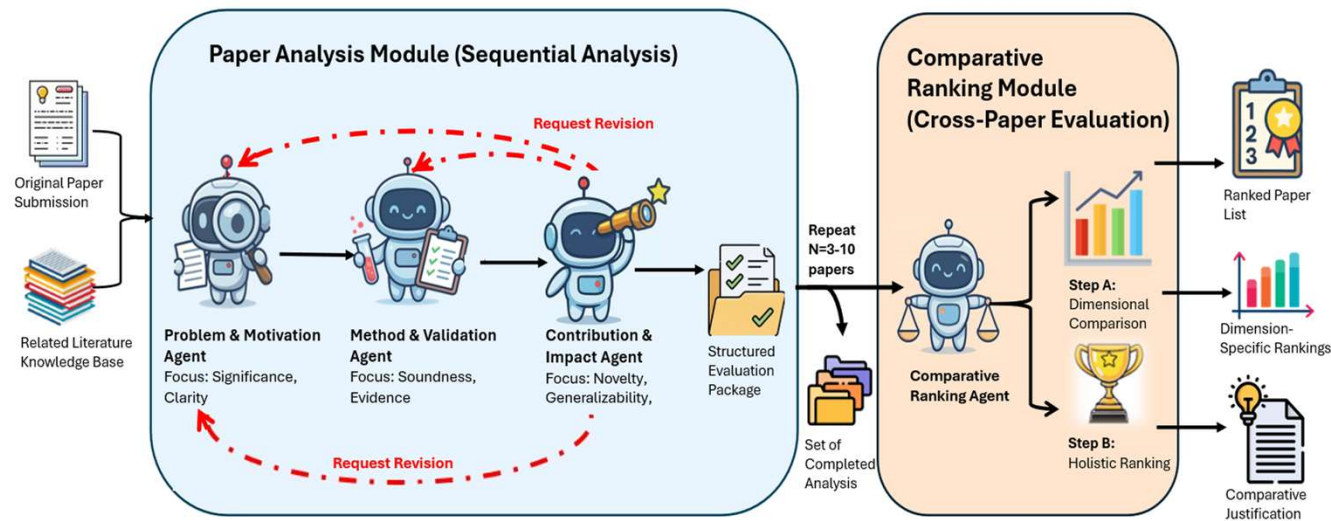


Figure 1: End-to-end ProComp workflow illustrating paper ingestion, specialized agent analyses, meta-review synthesis, and final score prediction

RESULTS

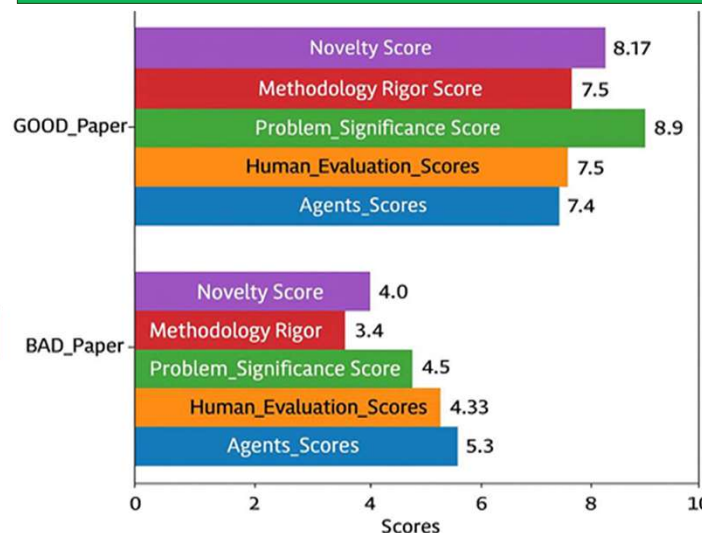


Figure 2: Comparison of GOOD and BAD paper scores across five evaluation dimensions, showing consistent quality separation

System	Category	n	MAE ↓	RMSE ↓
Main Models				
GPT	Main Model	194	0.9744	1.2239
NVIDIA	Main Model	194	1.2971	1.5394
QWEN	Main Model	194	1.4690	1.8042
LLAMA	Main Model	194	1.5629	1.8871
Baselines				
sea_c	Baseline	194	1.1490	1.4778
cycle_reviewer	Baseline	194	1.2132	1.5671
treereview	Baseline	194	1.2211	1.5684
agentreview	Baseline	194	1.2600	1.6179
deep_reviewer	Baseline	194	1.2735	1.6371
zero_shot	Baseline	194	1.3184	1.6982
chain_of_thought	Baseline	194	1.4543	1.8430

Table 1: Comparison of main models and baselines for overall score prediction (n=194). GPT shows the best alignment with human ratings.